✅ 1. Competency #7: Manipulate numbers, graphical and statistical data, algebraic expressions and equations, and geometric representations

- Highest relevance.
- Students must:
  - Treat the irregular oil slick as a quantifiable geometric object;
  - Select and apply an area estimation strategy (e.g., approximate it as a circle or ellipse, or use grid counting);
  - Measure lengths on the map (e.g., diameter = 4 cm);
  - Calculate map area (e.g., $\pi \times (2)^2 \approx 12.56$ cm²);
  - Apply the scale for area conversion:

$$1\ \text{cm} = 10\ \text{km} \Rightarrow 1\ \text{cm}^2 = (10\ \text{km})^2 = 100\ \text{km}^2$$

- Compute real-world area (e.g., 12.56 × 100 = 1256 km²).
- This entire process involves geometric modeling, numerical manipulation, and proportional reasoning, making it a quintessential example of Competency #7.

✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Solving requires a clear sequence of steps. For example, using the "approximate as a circle" method:
  1. Observe the oil slick and decide a circular approximation is reasonable;
  2. Measure the diameter on the map (e.g., 4 cm);
  3. Compute map area: $\pi \times (2)^2 \approx 12.56$ cm²;
  4. Convert using area scale: 12.56 × 100 = 1256 km²;
  5. Round to a sensible estimate (e.g., "about 1300 km²").

  Alternatively, using the grid method:

  1. Overlay a grid on the map;
  2. Count full and partial squares;
  3. Sum to get total map area;
  4. Multiply by 100 to get km².
- Both approaches require structured, sequential executi ↓ fully aligning with Competency #12.

✅ 3. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process (e.g., working to a context-appropriate level of accuracy)

- Highly relevant.
- The task explicitly asks for an "estimate", not an exact value.
- Contextual understanding shapes key decisions:
  - The oil slick has a fuzzy, irregular boundary → high precision is neither possible nor necessary;
  - The scale itself is approximate → results should be reported with 1–2 significant figures (e.g., 1200 or 1300 km², not 1256.637...);
  - The method should be efficient and reasonable (e.g., no need for calculus; simple geometric approximation suffices).
- A student who attempts overly precise tracing or excessive decimal places misunderstands the modeling purpose. Context directly determines method choice and acceptable precision.

✅ 4. Competencies #4 / #6: Make mathematical diagrams, graphs, constructions or computing artifacts; and extract mathematical information from them

- Moderately relevant.
- Students must:
  - Interpret the map as a scaled geometric representation;
  - Possibly add auxiliary marks (e.g., draw a best-fit circle, label diameter) or overlay transparent grid paper;
  - Extract dimensional or area-related information from the visual image.
- Although no new diagram is required, the act of transforming the map into a measurement tool demonstrates active engagement with mathematical representations—core to Competencies #4 and #6.

✅ 5. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Implicitly expected.
- While the prompt doesn't explicitly say "show your work," a credible estimate typically requires method justification, such as:
  - "I approximated the slick as a circle with a 4 cm diameter..."
  - "The map area is about 12 cm^2, so 12 × 100 = 1200 km^2."
- An isolated number without reasoning is hard to validate. Thus, clear communication of the estimation logic is implicitly valued, aligning with Competency #8.

✅ 6. Competency #1: Perform a simple calculation

- Basic but necessary.
- Involves squaring, multiplication, and approximating π (e.g., as 3.14 or 3). The arithmetic is straightforward.
- The key challenge lies in correctly applying the area scaling principle (area ratio = square of length ratio), not computational difficulty.

✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 学生必须：
  - 将不规则油污区域视为一个可量化的几何对象；
  - 选择并应用一种面积估算策略（如将其近似为圆、椭圆，或用方格覆盖计数）；
  - 测量图上长度（如直径 d cm），计算图上面积（如 $\pi(d/2)^2$ cm²）；
  - 运用比例尺进行面积单位换算：

$$1\ \text{cm} = 10\ \text{km} \Rightarrow 1\ \text{cm}^2 = (10\ \text{km})^2 = 100\ \text{km}^2$$

- 最终得出实际面积（如 12 cm² × 100 = 1200 km²）。
- 整个过程涉及几何建模、数值操作与比例推理，是能力 #7 的典型体现。

✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 解题需清晰的多阶段流程，例如（以“近似为圆”为例）：
  1. 观察油污形状，判断是否可近似为圆形；
  2. 用尺子测量图上直径（如 4 cm）；
  3. 计算图上面积：$\pi \times (2)^2 \approx 12.56$ cm²；
  4. 应用面积比例：12.56 × 100 = 1256 km²；
  5. 四舍五入或给出合理估算（如“约 1300 km²”）。
- 若采用方格法，则步骤为：覆盖网格 → 数完整/部分方格 → 求和 → 换算。
- 无论哪种方法，都要求结构化、有序的多步执行，符合能力 #12。

✅ 3. 能力 #13：利用对情境的理解来指导或加速数学解题过程（例如采用符合情境的精度）

- 高度相关。
- 题目关键词是“估算”（estimate），而非“精确计算”。
- 情境理解引导关键决策：
  - 石油扩散边缘模糊、形状不规则 → 无需追求高精度；
  - 比例尺本身是近似值 → 结果保留1–2 位有效数字即可（如 1200 或 1300 km²，而非 1256.637…）；
  - 方法选择应高效且合理（如不必要用积分，而可用简单几何近似）。
- 学生若试图“精确描边”或过度计算，反而违背了现实建模精神。情境意识直接决定了方法的合理性与精度水平。

✅ 4. 能力 #4 / #6：制作/使用数学图表并从中提取数学信息

- 中等相关。
- 学生需：
  - 解读地图作为比例缩放的几何表示；
  - 可能在图上添加辅助线（如画圆、标直径）或覆盖透明方格纸；
  - 从视觉图像中提取尺寸或面积信息。
- 虽然不强制制作新图，但主动将地图转化为测量工具的过程，体现了对数学图表的深度使用（#4 和 #6）。

✅ 5. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 隐含要求。
- 尽管题目未明确说“展示过程”，但在评估中，合理的估算需伴随方法说明，例如：
  - “我将油污近似为一个直径 4 cm 的圆…”
  - “图上面积约 12 cm^2，乘以 100 得 1200 km^2。”
- 缺乏解释的孤立数字难以判断其合理性。因此，清晰沟通估算逻辑是隐含期望，契合能力 #8。

✅ 6. 能力 #1：执行简单计算

- 基础支撑。
- 涉及平方、乘法、π 的近似（如 3.14 或 3），但计算本身简单。
- 关键在于正确应用比例关系（面积比 = 长度比的平方），而非算术难度。